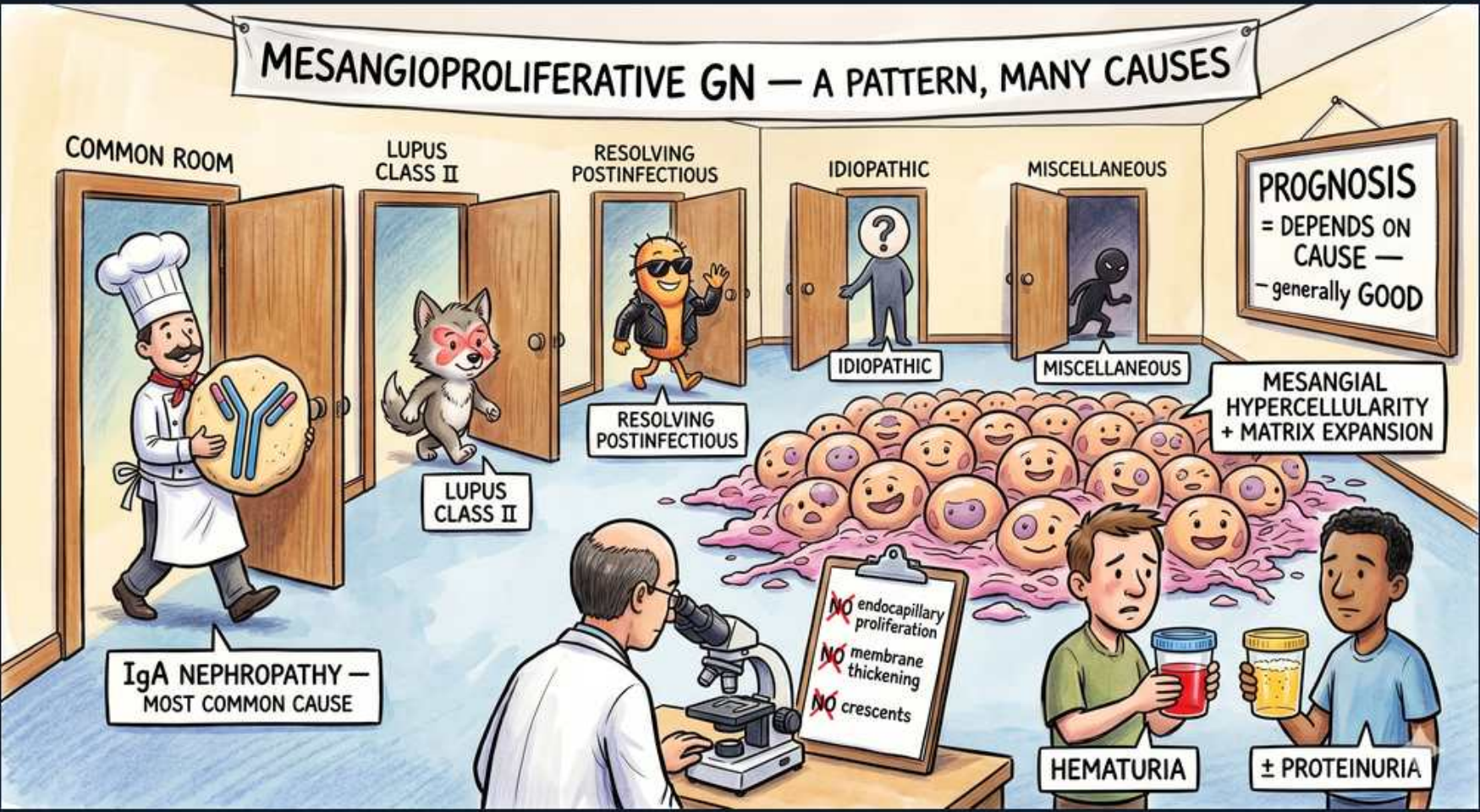


Mesangioproliferative GN — A Pattern, Many Causes



1. A pattern, many causes

WHAT YOU SEE

Title banner across the top

WHAT IT ENCODES

Mesangioproliferative GN is a histologic pattern — mesangial hypercellularity and matrix expansion — not a single disease. Defined by ≥3 mesangial cells per area on LM. The differential is driven by what the IF/EM deposits show.

BOARD TRAP

Reporting 'mesangioproliferative GN' without identifying the underlying cause is incomplete — always characterize the deposits and clinical context.

2. Mesangial hypercellularity + matrix expansion

WHAT YOU SEE

Right banner 'mesangial hypercellularity + matrix expansion'

WHAT IT ENCODES

The defining lesion: increased mesangial cell number and expanded mesangial matrix, expanding the mesangium on PAS. Capillary loops remain open (vs endocapillary/MPGN patterns). Deposits localize to the mesangium on EM.

BOARD TRAP

Mesangial-only proliferation (open loops) differs from MPGN's double contours and endocapillary GN's loop-filling proliferation — restricting to the mesangium is the key feature.

3. IgA nephropathy — most common cause

WHAT YOU SEE

'IgA nephropathy — most common cause' sign, lower left

WHAT IT ENCODES

IgA nephropathy is the most common cause of a mesangioproliferative pattern: mesangial IgA deposits on IF, mesangial EM deposits, presenting with synpharyngitic gross hematuria or microhematuria.

BOARD TRAP

A mesangioproliferative biopsy with DOMINANT mesangial IgA = IgA nephropathy — the IF immunoglobulin class makes the diagnosis.

4. Lupus Class II

WHAT YOU SEE

Wolf door labeled 'Lupus Class II'

WHAT IT ENCODES

ISN/RPS Class II lupus nephritis = mesangial proliferative lupus: mesangial hypercellularity with mesangial immune deposits ('full house' IF: IgG, IgA, IgM, C3, C1q). Mild disease, good prognosis, usually no aggressive immunosuppression.

BOARD TRAP

Class II (mesangial) is benign and treated supportively — proliferative Class III/IV with subendothelial deposits is what needs cyclophosphamide/MMF.

5. Resolving post-infectious GN

WHAT YOU SEE

Door 'resolving postinfectious'

WHAT IT ENCODES

In the healing phase of post-infectious (post-strep) GN, the florid endocapillary proliferation resolves into a residual mesangioproliferative pattern with normalizing complement — a recovering look-alike.

BOARD TRAP

A mesangioproliferative pattern weeks after pharyngitis with NORMALIZING C3 is resolving PIGN, not a new chronic GN — timing of complement recovery distinguishes.

6. Idiopathic

WHAT YOU SEE

Door with '?' labeled idiopathic

WHAT IT ENCODES

Idiopathic mesangioproliferative GN: a pattern without an identifiable cause or dominant Ig (sometimes IgM nephropathy or C1q nephropathy). Diagnosis of exclusion after ruling out IgA, lupus, and infection.

BOARD TRAP

Calling a case idiopathic before excluding IgA/lupus/infection and checking IF for IgM or C1q is premature — exclusion is mandatory.

7. Miscellaneous / secondary

WHAT YOU SEE

Dark door labeled 'miscellaneous'

WHAT IT ENCODES

Other secondary causes producing mesangial proliferation include IgA vasculitis (HSP), resolving other GNs, and early/mild forms of various immune-complex diseases. The pattern is a common final pathway.

BOARD TRAP

The pattern is nonspecific — anchoring on one diagnosis without IF/serology risks missing a systemic disease like lupus or vasculitis.

8. No endocapillary / no crescents

WHAT YOU SEE

Negated cards: endocapillary proliferation, membrane thickening, crescents

WHAT IT ENCODES

Defining the pattern by what's ABSENT: no endocapillary (loop-filling) proliferation, no GBM thickening/double contours, and no crescents. Pure mesangial expansion with patent capillaries = the milder mesangioproliferative end of the spectrum.

BOARD TRAP

Adding crescents or endocapillary proliferation moves you to crescentic or proliferative (Class IV/MPGN) disease — those carry worse prognosis and need immunosuppression.

9. Hematuria

WHAT YOU SEE

Hematuria red urine cup, lower center

WHAT IT ENCODES

Clinically presents with glomerular hematuria (dysmorphic RBCs, RBC casts) — often microscopic, sometimes gross with intercurrent infection (IgA pattern). The nephritic signature of mesangial injury.

BOARD TRAP

Glomerular vs urologic hematuria: dysmorphic RBCs and RBC casts confirm a glomerular source — isomorphic RBCs suggest a urologic cause.

10. ± Proteinuria

WHAT YOU SEE

± proteinuria yellow cup, lower right

WHAT IT ENCODES

Proteinuria is usually mild and sub-nephrotic in pure mesangioproliferative disease. Heavy/nephrotic proteinuria suggests a more aggressive lesion (FSGS transformation, proliferative lupus) rather than simple mesangial proliferation.

BOARD TRAP

Nephrotic-range proteinuria argues AGAINST bland mesangioproliferative GN — re-examine for FSGS or a proliferative class lesion.

11. Prognosis depends on cause

WHAT YOU SEE

'Prognosis = depends on cause — generally good' banner

WHAT IT ENCODES

Because the pattern spans benign (Class II lupus, resolving PIGN) to progressive (proteinuric IgA) entities, outcome is driven by the underlying cause and severity — generally good for pure mesangial disease.

BOARD TRAP

There's no single prognosis for 'mesangioproliferative GN' — outcome tracks the etiology, so identifying the cause is the prognostic step.
